

CSC 335 Assignment Report

Building a "My Profile" Page with Git History

This laboratory report presents the complete practical implementation of the CSC 335 Profile Project assignment.

The primary goal was to construct a multi-page, minimalist personal web profile while rigorously executing

and documenting fundamental and advanced version control workflows with Git and GitHub.

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Department: Computer Science (300L)

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GitHub Repository: <https://github.com/Achiever864/profile-project>

- * Repository initialization, iterative semantic commits, and remote tracking on GitHub.
 - * Feature branch lifecycle management, fast-forward merges, and non-linear branch topologies.
 - * External CSS design system integration and unified multi-page responsive navigation.
 - * Deliberate merge conflict generation, manual conflict resolution using standard markers, and resolution commits.
- ### 1. Executive Summary & Assignment Objectives
- * Staging area inspection via `git diff` and `git diff --staged`, alongside `.gitignore` pattern verification.
 - * Multi-feature implementation culminating in an Academic Dashboard (`dashboard.html`) and comprehensive Git tree analysis.

2. Registered Semester Courses (300L Harmattan Semester)

- MAT352:** Numerical Analysis II (3 Units, Compulsory) — Numerical solutions of ODEs, eigenvalue problems, error analysis
- CSC301:** Systems Analysis & Structured Programming (3 Units, Compulsory) — Data structures, algorithm design, modeling, modular programming, encryption
- CSC313:** Data Structures & Algorithms (3 Units, Compulsory) — Graphs, hashing, asymptotic analysis, greedy algorithms, dynamic programming
- CSC335:** Web Application Development (3 Units, Compulsory) — Front-end web architecture, HTML5/CSS3, Git workflows, client-server interaction
- GES103:** Government, Society and Economy (2 Units, Required) — Political science dynamics, socio-economic structures, civic responsibilities
- GES301:** Introduction to Entrepreneurship Skills (2 Units, Required) — Business plan development, opportunity recognition, enterprise leadership

3. Rendered Profile Webpage Preview (profile.html)



Phase 1: Repository Setup, Course Listing, & Branch Merging (Tasks 1 - 11)

- Task 1: Repository profile-project initialized with Git tracking.
- Task 2 & 3: Created profile.html containing student personal details (Igaga Ikhuemoisa Adeoluwa, 243457, Computer Science, 300L) and interest statement. Staged and committed with message:

'Initial commit: create profile page'.

- Task 4: Appended Current Courses section to profile.html. Staged and committed with message: 'Add list of current courses'.

- Task 5: Ran git log --oneline to verify sequential commits:

```
PowerShell - Task 5: git log --oneline

$ git log --oneline
c4749f8 Add list of current courses
0540d13 Initial commit: create profile page
```

- Task 6: Created and checked out branch design-update. Applied clean CSS typography and background styling. Committed with: 'Update background colour and typography styling'.

Task 7: Switched back to main branch; (reverted profile.html) and verified that styling was absent on main.

- Task 8: Merged design-update into main (fast-forward merge), verifying that design styles now appear.
- Task 9: Executed git log --oneline to confirm updated history:

```
PowerShell - Task 6 to 8: design-update Branching & Merge

$ git checkout -b design-update
Switched to a new branch 'design-update'
$ git commit -m 'Update background colour and typography styling'
[design-update 173ae0e] Update background colour and typography styling
1 file changed, 14 insertions(+)
$ git checkout main
Switched to branch 'main'
# Verified: profile.html on main has no design changes
$ git merge design-update
Updating c4749f8..173ae0e
Fast-forward: profile.html | 14 ++++++
```

```
PowerShell - Task 9: git log --oneline after Merge

$ git log --oneline
173ae0e Update background colour and typography styling
c4749f8 Add list of current courses
0540d13 Initial commit: create profile page
```

- Task 10: Configured remote origin: <https://github.com/Achiever864/profile-project> and pushed via git push -u origin main.

Task 11: Created repository on GitHub and pushed profile-project to remote origin. Verified commit log and files match identically.

Tasks 10 & 11: Remote GitHub Integration & Clone Verification

```
PowerShell - Task 10 & 11: Remote Linking & Clone Verification

$ git remote -v
origin https://github.com/Achiever864/profile-project (fetch)
origin https://github.com/Achiever864/profile-project (push)
$ git clone https://github.com/Achiever864/profile-project profile-project-check
Cloning into 'profile-project-check'...
$ cd profile-project-check && git log --oneline
173ae0e Update background colour and typography styling
c4749f8 Add list of current courses
0540d13 Initial commit: create profile page
# Verified: Cloned repository byte-for-byte matches local project
```

Task 12 Answers & Multi-Page Architecture (Tasks 12 - 18)

Task 12: Conceptual Analysis of Git Fundamentals (Written Answers)

Question A: What is the difference between 'git add' and 'git commit'?

- 'git add' interacts with the Staging Area (Index). When files are modified or created in the working directory, Git does not track them automatically. 'git add' stages specific changes, giving developers precise control over what modifications will be included in the next snapshot.
- 'git commit' permanently records the staged snapshot into the local Git repository history. It creates a cryptographic commit object (SHA-1/SHA-256 hash) with metadata (author, timestamp, parent commit, and commit message). Without 'git commit', changes remain ephemeral in the staging area.

Question B: What is the difference between 'git push' and 'git pull'?

- 'git push' uploads local branch commits to a designated remote repository (e.g., GitHub). It updates the remote branch reference, publishing local work so collaborators can access it.
- 'git pull' downloads changes from a remote tracking branch and immediately integrates them into the currently checked-out local branch. Mechanically, 'git pull' is a compound command equivalent to running 'git fetch' (downloading remote objects) followed by 'git merge' (combining branches).

Tasks 13 - 15: courses.html, External CSS (css/style.css) & Site Navigation

- Task 13: Created courses.html with comprehensive descriptions for MAT352, CSC301, CSC313, CSC335, GES103, and GES301. Committed with: 'Add courses.html with course codes and descriptions'.
- Task 14: Created css/style.css and extracted all inline and embedded styling into a centralized, modern design system featuring CSS variables (--bg-main, --card-bg, --accent, --border-color).
- Task 15: Implemented a responsive navigation bar (<header><nav>) linking Home/Profile and Courses across both pages. Committed with: 'Extract styles to external css/style.css and add navigation menu'.

- Task 16 & 17: Created branch about-page. Developed about.html containing:
 - Short Biography: Educational background and computing philosophy.
 - Academic Interests: Software Architecture, Algorithmic Optimization, Distributed Cloud Systems.
 - Career Goals: Aspiring Lead Software Architect and Full-Stack Systems Engineer.

Tasks 16 - 18: about-page Branch, about.html, & Branch Integration

- Technical Toolkit: Python, JavaScript, C++, Git, Docker.
- Styled using css/style.css and committed on about-page branch.
- Task 18: Switched to main branch, merged about-page (git merge about-page), and pushed to remote origin. Verified about.html is fully accessible and tracked on main.

Phase 3: Forms, .gitignore & Merge Conflict Resolution (Tasks 19 - 22)

Tasks 19 & 20: contact.html & .gitignore Configuration with Ignored Files

- Task 19: Built contact.html with student contact details and interactive form (Name, Email, Message, Submit).
- Task 20: Created .gitignore with ignore patterns for notes.txt, *.log, and temporary build files.
Created test files notes.txt and debug.log in the project directory. Executed git status to confirm Git ignores them:

```
PowerShell - Task 20: .gitignore Status Verification

$ cat .gitignore
notes.txt
*.log
evidence/
*.tmp
$ New-Item notes.txt, debug.log
$ git status
On branch main
Untracked files:
  .gitignore
# Notice: notes.txt and debug.log are completely ignored by Git!
```

- Task 21: Created branch profile-update. Modified software engineering interest paragraph in profile.html. Committed on profile-update: 'Update software engineering statement on profile-update branch'.
 - Task 22: Switched to main and edited the EXACT same paragraph with competing content. Committed on main: 'Update software engineering statement on main branch'.
- Attempted git merge profile-update, successfully producing CONFLICT (content). Merge conflict in profile.html.

Tasks 21 & 22: Merge Conflict Simulation, Conflict Markers & Manual Resolution

```
PowerShell - Task 21 & 22: Merge Conflict Simulation in profile.html

$ git checkout profile-update
$ git commit -m 'Update software engineering statement on profile-update branch'
[profile-update cfdeb59] Update software engineering statement on profile-update branch
$ git checkout main
$ git commit -m 'Update software engineering statement on main branch'
[main 5e3dbce] Update software engineering statement on main branch
$ git merge profile-update
Auto-merging profile.html
CONFLICT (content): Merge conflict in profile.html
Automatic merge failed; fix conflicts and then commit the result.
```

```
profile.html - Git Conflict Markers (Task 22)

49: <section id="interest">
50:   <h2>Software Engineering Interest</h2>
51: <==== HEAD
52:   <p>Main branch perspective: I am deeply invested in software
    engineering with emphasis on algorithmic efficiency...</p>
53: =====
54:   <p>Branch profile-update perspective: I am dedicated to modern
    software engineering, focusing specifically on microservices...</p>
55: >>>>>> profile-update
56: </section>
```

```
PowerShell - Task 22: Conflict Resolution & Completed Merge

# Resolved by harmonizing both perspectives cleanly in profile.html
$ git add profile.html
$ git commit -m 'Resolve merge conflict in profile.html between main and profile-update'
[main 5e61ca7] Resolve merge conflict in profile.html between main and profile-update
$ git push origin main
To https://github.com/Achiever864/profile-project
5e3dbce..5e61ca7 main -> main
```

Phase 4: Git Diff Analysis & Academic Dashboard (Tasks 23 - 24)

- Comparative Analysis:
- 'git diff': Compares the Working Directory against the Staging Area (Index). It displays line-level edits that have been made to tracked files but have NOT yet been added to the staging index.
 - 'git diff --staged' (or 'git diff --cached'): Compares the Staging Area against the most recent commit (HEAD). It shows exactly what modifications will be committed in the next 'git commit'.
 - Operational Demonstration: Added a new academic interest to about.html. Before git add, git diff printed the added line while git diff --staged was blank. After git add, git diff was empty and git diff --staged displayed the staged diff ready for committing.

```
PowerShell - Task 23: git diff (Unstaged) vs git diff --staged

$ git diff
diff --git a/about.html b/about.html
--- a/about.html
+++ b/about.html
@@ -37,6 +37,7 @@
+ <li><strong>Computational Logic & Formal Methods:</strong> Automata...</li>
# Staging changes...
$ git add about.html
$ git diff # Returns EMPTY because working directory matches staging area
$ git diff --staged
diff --git a/about.html b/about.html
--- a/about.html
+++ b/about.html
@@ -37,6 +37,7 @@
+ <li><strong>Computational Logic & Formal Methods:</strong> Automata...</li>
```

- Created dedicated feature branch dashboard-feature.
 - Developed dashboard.html containing:
 - Student credentials (Igaga Ikhuemoisa Adeoluwa, Matric: 243457, Computer Science, 300L).
 - Course Registration Table: Complete course roster with codes, titles, status, and credit units (16 Units Total).
- Academic Summary: First Class division standing, session details, and computational concentration.
- Task 24: Final Feature Branch (dashboard-feature) & Academic Dashboard (dashboard.html)**
- Committed on dashboard-feature, pushed branch to GitHub, merged into main, and pushed final main branch to remote.

https://achiever864.github.io/profile-project/dashboard.html

Profile Project

Home/Profile | Courses | About | Contact | Dashboard

Academic Dashboard

Comprehensive overview of registered courses, credit units, and academic summary

Student Overview

Student Name: Igaga Ikhuemoisa Adeoluwa | Matriculation: 243457
Faculty of Science | Department: Computer Science (300L)

Course Registration Table (16 Units)

MAT352 (3 Units), CSC301 (3 Units), CSC313 (3 Units)
CSC335 (3 Units), GES103 (2 Units), GES301 (2 Units) ---> Total Units: 16

Academic Summary

Session: 2025/2026 Harmattan Semester | Standing: First Class Division
Core Focus: Distributed Architecture, Algorithms, & Cloud Systems

Task 25: Git Graph & Written Conceptual Questions

Task 25A: Complete Git Commit Tree (git log --oneline --graph --all)

```
PowerShell - Task 25: git log --oneline --graph --all

$ git log --oneline --graph --all
* a62f0a8 Create dashboard.html with course units table and academic summary
* ef0c577 Update academic interests in about.html and demonstrate git diff
* 5e61ca7 Resolve merge conflict in profile.html between main and profile-update
| \
| * cfdeb59 Update software engineering statement on profile-update branch
* | 5e3dbce Update software engineering statement on main branch
| /
* 122d72a Add .gitignore to ignore notes.txt and log files
* 0eb0a9d Add contact.html with contact form and details, and update site nav
* e0ff1ff Create about.html with bio and apply external styles
* 2792771 Extract styles to external css/style.css and add navigation menu
* b7997e0 Add courses.html with course codes and descriptions
* 173ae0e Update background colour and typography styling
* c4749f8 Add list of current courses
* 0540d13 Initial commit: create profile page
```

Question 4: What is the difference between a local and remote repository?

Answer: A local repository resides on the developer's personal machine (in the hidden .git directory), enabling complete offline version control—including branching, staging, committing, and viewing history. A remote repository is hosted on an external server or cloud platform (such as GitHub, GitLab, or Bitbucket) and acts as a centralized synchronization hub for backup, code review, continuous integration, and team collaboration.

Task 25B: Answers to Written Conceptual Questions